Fuzzy Regression Discontinuity

And a working review of Instrumental Variables

ECO372H1F — Data Analysis & Applied Econometrics in Practice

University of Toronto • Winter 2026

What this lecture is for

We have now seen two designs that each recover a LATE:

- **Sharp RD** (Lecture 4): a cutoff rule assigns treatment deterministically. Continuity at the cutoff buys us the effect *at* the cutoff.
- **IV** (Lecture 6): a clever third variable moves the endogenous regressor. Relevance + exclusion buy us the effect *for compliers*.

A fuzzy RD design combines the two: it is an IV design in which the instrument is "the unit landed on the treated side of the cutoff."

Goal

By the end you should be able to set up, estimate, defend, and interpret a fuzzy RD design, and explain why the cutoff indicator serves as an instrument for treatment received.

Roadmap

1. IV review: the four assumptions, the Wald ratio, compliers.
2. From sharp to fuzzy: imperfect compliance at the cutoff.
3. **The key idea: the cutoff indicator is an instrument.**
4. The fuzzy estimand $\tau_{FRD} = \tau_Y / \tau_D$ and why the naive jump is biased toward zero.
5. Response types translated into RD language; one-sided non-compliance.
6. Estimation in practice (`rdrobust`, bandwidth, SEs).
7. Threats: continuity, exclusion, monotonicity, weak first stage.
8. Worked example: the last-chance exam (the sheepskin effect).

IV review

The problem IV solves

We believe a causal model

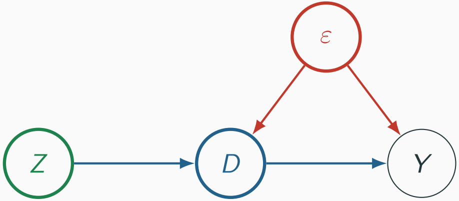
$$Y_i = \alpha + \beta D_i + \varepsilon_i,$$

but D_i is endogenous: $\mathrm{Cov}(D_i, \varepsilon_i) \neq 0$ (omitted variables, measurement error, or simultaneity). Then OLS is biased and *more data does not help*:

$$\text{plim}\, \hat{\beta}_{OLS} = \beta + \frac{\mathrm{Cov}(D_i, \varepsilon_i)}{\mathrm{Var}(D_i)}.$$

The fix. Do not use all the variation in D. Find a variable Z that moves D but is otherwise clean, and use *only* that slice of variation.

The instrument, as a picture



A valid instrument Z needs:

- **Relevance.** $\mathrm{Cov}(Z, D) \neq 0$.
  (arrow $Z \to D$ exists)
- **Independence.** Z unrelated to ε.
  (no arrow $\varepsilon \to Z$)
- **Exclusion.** Z touches Y *only* through D.
  (no arrow $Z \to Y$)

Slogan

Exogenous variation in Z changes the endogenous D; any resulting change in Y reflects only the causal effect of D.

The Wald estimator

Take $\mathrm{Cov}(\cdot, Z_i)$ of the structural equation. Exclusion kills the error term, relevance lets us divide:

$$\beta = \frac{\mathrm{Cov}(Z_i, Y_i)}{\mathrm{Cov}(Z_i, D_i)}.$$

With a binary instrument this is just four means — the **reduced form** over the **first stage**:

$$\beta_{IV} = \frac{\overbrace{E[Y \mid Z = 1] - E[Y \mid Z = 0]}^{\text{reduced form } = \rho}}{\underbrace{E[D \mid Z = 1] - E[D \mid Z = 0]}_{\text{first stage } = \pi}} = \frac{\rho}{\pi}.$$

The reduced form is the effect of the *offer* (the intent-to-treat, ITT). Dividing by the first stage rescales "effect of being offered" into "effect of actually taking it." This ratio is the basis for the fuzzy RD estimator developed below.

Heterogeneous effects $\Rightarrow$ LATE

Effects differ across people: each i has its own $\beta_i = Y_i(1) - Y_i(0)$. With a binary instrument, sort people by their potential treatments $(D_i(0), D_i(1))$:

	$D_i(0) = 0$	$D_i(0) = 1$
$D_i(1) = 1$	**Complier**	Always-taker
$D_i(1) = 0$	Never-taker	Defier

- **Always-taker**: takes treatment regardless of Z.
- **Never-taker**: never takes it, regardless of Z.
- **Complier**: takes it *iff* the instrument says so.
- **Defier**: does the opposite. Ruled out by assumption.

The four IV assumptions (heterogeneous version)

1. **Independence.** $\{Y_i(d,z), D_i(z)\} \perp\!\!\!\perp Z_i$. ($Z$ as-good-as-random.)
2. **Exclusion.** $Y_i(d,1) = Y_i(d,0)$. (Z acts on Y only through D.)
3. **First stage.** $E[D_i(1) - D_i(0)] \neq 0$. (Relevance, and it is *testable*.)
4. **Monotonicity.** $D_i(1) \geq D_i(0)$ for all i. (No defiers.)

The LATE theorem (Imbens & Angrist, 1994)

Under all four,

$$\beta_{IV} = E\big[\, Y_i(1) - Y_i(0) \,\big|\, \text{complier} \,\big].$$

IV recovers the average effect for compliers — the people the instrument actually moved. A different instrument $\to$ a different complier group $\to$ a different LATE.

Why monotonicity matters, and type shares

Monotonicity (no defiers) leaves only three types, and lets us *count* them even though we can never label an individual:

$$\begin{aligned} P(\text{always-taker}) &= P(D = 1 \mid Z = 0), \\ P(\text{never-taker}) &= P(D = 0 \mid Z = 1), \\ P(\text{complier}) &= P(D = 1 \mid Z = 1) - P(D = 1 \mid Z = 0) \ = \ \text{the first stage}. \end{aligned}$$

A bigger first stage $\Rightarrow$ more compliers $\Rightarrow$ the LATE speaks for a larger share of the population. This is exactly why we report and scrutinise the first stage.

From sharp to fuzzy

Sharp RD: a quick recap

A rule assigns treatment based on a running variable X crossing a cutoff c:

$$D_i = \mathbf{1}\{X_i \geq c\}.$$

Just below vs. just above the cutoff is as-good-as-random, so the jump in the outcome at c is the treatment effect:

$$\tau_{SRD} = \lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x] = E[Y(1) - Y(0) \mid X = c].$$

The identifying assumption is continuity: absent treatment, $E[Y(0) \mid X]$ and $E[Y(1) \mid X]$ would pass smoothly through c. Probability of treatment jumps cleanly from **0 to 1**.

Fuzzy RD: the probability of treatment jumps, not treatment

In a fuzzy design, crossing the cutoff changes the *probability* of treatment — discontinuously, but not from 0 to 1:

$$\lim_{x \downarrow c} P(D = 1 \mid X = x) \; > \; \lim_{x \uparrow c} P(D = 1 \mid X = x), \quad \text{jump } < 1.$$

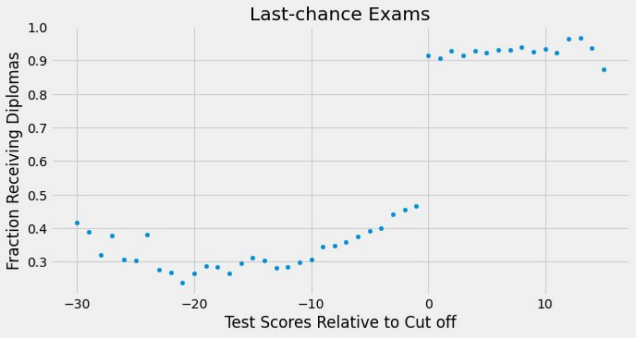
This is **imperfect compliance**. Two things can happen:

- Some units above the cutoff are *not* treated (never-takers).
- Some units below the cutoff *are* treated (always-takers).

Fuzziness is the rule, not the exception, in real policy:

- Legal drinking age: turning 21 raises, but does not guarantee, drinking.
- Scholarship eligibility $\neq$ take-up.
- Benefit cutoffs: eligible $\neq$ enrolled.

The fuzzy first stage you can see



Texas “last-chance” exams: passing the cutoff is supposed to grant a diploma. Data: Clark & Martorell (2014), via Facure, *Causal Inference for the Brave and True*, ch. 16.

The treatment is “received a diploma”; the running variable is the test score relative to the pass cutoff.

Notice the first stage is fuzzy: the diploma probability does not go $0 \to 1$. It jumps from roughly **0.5 to 0.9**.

Below the cutoff some students get a diploma anyway (**always-takers**); above it some do not (**never-takers**).

Two pictures that define the design

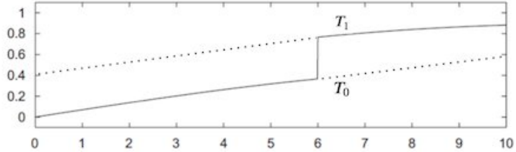


Fig. 3. Assignment probabilities (FRD).

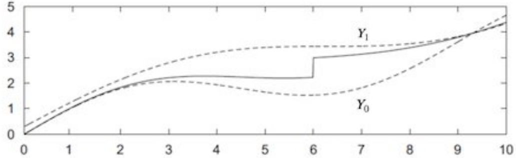


Fig. 4. Potential and observed outcome regression (FRD).

Imbens & Lemieux (2008), *RD designs: a guide to practice*, Figs 3–4. Cutoff at $X = 6$.

Top — assignment probabilities (the first stage). T_1 is the treatment probability you would have just above the cutoff, T_0 just below. At $c = 6$ they *jump apart* but neither hits 0 or 1. The size of that gap is τ_D.

Bottom — outcomes. Dashed curves are the potential outcomes Y_0, Y_1 (smooth through c — continuity). The solid *observed* curve jumps at c because the treated *share* jumps. That outcome jump is τ_Y.

The effect we want is the outcome jump *rescaled* by the probability jump.

The key idea: the cutoff is an instrument

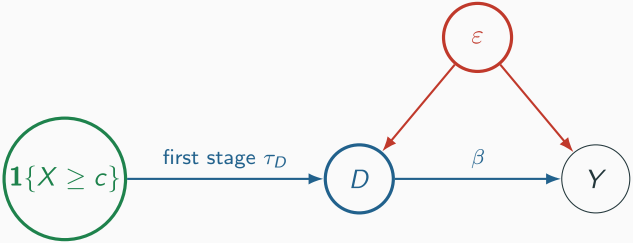
Line up the two designs

The fuzzy first stage looks exactly like *non-compliance with an offer*. So treat "you crossed the cutoff" as a randomly assigned offer, i.e. an instrument.

IV object	**Symbol**	**Fuzzy RD counterpart**
Instrument Z	$Z_i = \mathbf{1}\{X_i \geq c\}$	landed on the treated side
Endogenous regressor D	D_i	treatment actually received
Outcome Y	Y_i	outcome
First stage	π	jump in $P(D{=}1)$ at c (τ_D)
Reduced form / ITT	ρ	jump in $E[Y]$ at c (τ_Y)

Near the cutoff, Z_i is as-good-as-random (continuity / no sorting), which is precisely the **independence** you need for an instrument.

The fuzzy-RD DAG is the IV DAG



- **Relevance:** crossing the cutoff really shifts D — the visible probability jump.
- **Independence:** near c, which side you land on is as-good-as-random.
- **Exclusion:** crossing the cutoff affects Y *only* by changing D — not through any other channel.

The fuzzy estimand

Because the cutoff indicator is the instrument, the fuzzy RD effect *is* a Wald ratio — reduced form over first stage, taken in the limit at the cutoff:

$$\boxed{\tau_{FRD} = \frac{\tau_Y}{\tau_D} = \frac{\lim_{x \downarrow c} E[Y \mid X{=}x] - \lim_{x \uparrow c} E[Y \mid X{=}x]}{\lim_{x \downarrow c} E[D \mid X{=}x] - \lim_{x \uparrow c} E[D \mid X{=}x]}}$$

Under continuity + monotonicity this identifies

$$\tau_{FRD} = E\big[\, Y_i(1) - Y_i(0) \;\big|\; \text{complier},\; X_i = c \,\big].$$

Doubly local

The estimate is local in *two* senses: at the cutoff (the RD locality) *and* for compliers (the IV locality).

Why the naive (sharp) jump understates the effect

Suppose you ignored the fuzziness and just read off the outcome jump τ_Y as if it were the treatment effect. You would be biased toward zero.

Above the cutoff, not everyone is treated, so the observed average sits *below* the true Y_1 curve. Below the cutoff, some are treated, so the observed average sits *above* the true Y_0 curve. The observed gap is therefore *smaller* than the true treatment-effect gap.

Dividing by the first stage $\tau_D < 1$ *inflates* the gap back to the per-complier effect:

$$\tau_Y = \tau_D \times \tau_{FRD} \quad \Longrightarrow \quad \tau_{FRD} = \frac{\tau_Y}{\tau_D}.$$

Same logic as ITT $\div$ first stage in the lottery example. If compliance were perfect ($\tau_D = 1$) we are back to sharp RD.

Response types, in RD language

With $Z_i = \mathbf{1}\{X_i \geq c\}$, the same four types reappear:

- **Always-takers:** treated on *both* sides of the cutoff. (e.g. a student who gets the diploma even after failing.)
- **Never-takers:** untreated on *both* sides. (passes the exam but still gets no diploma.)
- **Compliers:** treated iff above the cutoff. These are who τ_{FRD} is about.
- **Defiers:** treated iff *below*. Assumed away by **monotonicity** — crossing the cutoff never *lowers* your chance of treatment.

The denominator τ_D *is* the complier share at the cutoff: the jump in treatment probability counts exactly the people whose treatment status flips when they cross.

One-sided non-compliance: a clean special case

Sometimes the rule makes treatment *impossible* on one side. If nobody below the cutoff can be treated, then

$$P(D = 1 \mid X < c) = 0 \quad \Longrightarrow \quad \text{there are no always-takers.}$$

This is **one-sided non-compliance**. The population near the cutoff is then just *compliers* + *never-takers*, and the first stage is

$$\tau_D = P(D = 1 \mid X \geq c) - 0 = P(\text{complier} \mid X = c).$$

This arises whenever eligibility is a strict precondition for treatment — a benefit that simply cannot be received without clearing the cutoff. Always-takers are eliminated by construction; defiers are already gone by monotonicity. The first stage then equals the complier share directly.

Estimation in practice

Two local regressions, then divide

Estimate the numerator and denominator with the *same* local-linear (or low-order polynomial) RD specification, recentering $X' = X - c$:

$$\textbf{Reduced form:} \quad Y_i = \rho_0 + \rho_1 \mathbf{1}\{X_i \geq c\} + f_Y(X_i') + v_i,$$

$$\textbf{First stage:} \quad D_i = \pi_0 + \pi_1 \mathbf{1}\{X_i \geq c\} + f_D(X_i') + \eta_i,$$

where f_Y, f_D are flexible functions allowed to differ on each side. Then

$$\hat{\tau}_{FRD} = \frac{\hat{\rho}_1}{\hat{\pi}_1} = \frac{\hat{\tau}_Y}{\hat{\tau}_D}.$$

Equivalently, run 2SLS of Y on D instrumented by $\mathbf{1}\{X \geq c\}$, with the running-variable polynomial as included controls.

Practical guidance

- **Plot first, always.** Show the first-stage jump and the reduced-form jump separately before reporting a ratio.
- **Use** `rdrobust` (Calonico–Cattaneo–Titiunik): pass the treatment-received variable as `fuzzy=`. It picks an optimal bandwidth and gives bias-corrected, robust CIs.
- **Bandwidth.** For a transparent ratio, use the *same* bandwidth in numerator and denominator.
- **Standard errors.** Do *not* read OLS SEs off a hand-run second stage — they ignore first-stage estimation. Use a fuzzy-RD command, the delta method, or bootstrap.
- **Clustering.** Cluster at the level at which the running variable / assignment varies (e.g. neighbourhood, school).
- **Polynomial order.** Stick to linear or quadratic; high-order polynomials overfit near the boundary (Gelman & Imbens 2019).

Reading the three numbers

A fuzzy RD report has three quantities; the third is the first two divided. Take an *illustrative* program offered to units above a cutoff, with imperfect take-up:

First stage	$\hat{\tau}_D = 0.50$	jump in P(treated) at the cutoff
Reduced form (ITT)	$\hat{\tau}_Y = 0.10$	jump in the outcome at the cutoff
Fuzzy RD	$\hat{\tau}_{FRD} = \dfrac{0.10}{0.50} = 0.20$	effect for compliers at the cutoff

Crossing the cutoff raised take-up by 50 p.p. and the outcome by 10 p.p.; since only half of those nudged actually took the treatment, the per-complier effect is twice the ITT. Report all three, and plot the first stage and reduced form before quoting the ratio.

Threats and diagnostics

What has to hold (and how to defend it)

Fuzzy RD inherits *both* RD and IV assumptions:

- **Continuity / no sorting.** Potential outcomes are smooth at c; units cannot manipulate X to pick a side. *Defend:* McCrary density test (`rddensity`), covariate balance at the cutoff, institutional knowledge.
- **Relevant first stage.** The treatment-probability jump τ_D is well separated from zero. *Defend:* plot it; report the first-stage F / p-value.
- **Exclusion.** Crossing the cutoff moves Y *only* through D. *Threat:* the mere *information* of being on the eligible side changes behaviour (signalling, morale). In the continuity framework this is implied by smoothness of the underlying conditional expectations.
- **Monotonicity.** No defiers; crossing the cutoff never reduces the chance of treatment. *Defend:* argue on institutional grounds.

Two failure modes worth naming

Weak first stage. A small τ_D is a small denominator: confidence intervals explode, and any slight exclusion violation is amplified,

$$\text{plim}\, \hat{\tau}_{FRD} = \tau_{FRD} + \frac{\mathrm{Cov}(Z, \varepsilon)}{\tau_D}.$$

Weak *and* slightly invalid is far worse than either alone.

Confounded cutoff. If a *different* program switches on at the same threshold, exclusion fails — the jump in Y mixes two treatments. (Classic example: Medicare *and* retirement norms both at age 65.) *Defend:* placebo outcomes that should not respond, and placebo cutoffs where no policy changes.

Worked example: the last-chance exam

The setting: does the diploma itself pay?

Two competing views of why education raises earnings:

- **Human capital:** schooling makes you more productive.
- **Signalling (the sheepskin effect):** the diploma certifies productivity you already had. What pays is the credential, not the extra learning.

Clark & Martorell (2014) separate these using a Texas exit exam. Students get several attempts, ending with a *last-chance* exam in 12th grade.

- Running variable: last-chance exam score, centered at the pass cutoff.
- Treatment D: receiving the diploma.
- Outcome Y: later earnings.

Students just below and just above the pass mark have near-identical human capital, but different credentials.

Why it is fuzzy, and the IV mapping

Passing is *supposed* to grant the diploma, but compliance is imperfect (recall the first-stage plot: the diploma probability jumps from roughly 0.5 to 0.9, not 0 to 1). So this is a fuzzy RD.

The cutoff indicator is the instrument:

- Instrument Z: $\mathbf{1}\{\text{score} \geq \text{pass cutoff}\}$.
- Endogenous regressor D: received the diploma.
- Outcome Y: earnings.
- **Exclusion (in words):** barely passing the exam affects earnings *only* by changing whether the student holds a diploma — not directly through, say, the test score itself signalling ability to employers.

Response types here

- **Always-takers:** receive the diploma even after failing the last-chance exam (other routes to graduation).
- **Never-takers:** pass but still do not get the diploma.
- **Compliers:** get the diploma *because* they passed. τ_{FRD} is their effect.
- **Defiers:** assumed away by monotonicity.

Because some failing students still graduate (always-takers) and some passing students do not (never-takers), the first stage is the gap between the two diploma probabilities — the complier share at the cutoff.

The result

The reduced form is the jump in earnings at the pass cutoff. In the data, earnings rise smoothly with the score but show no visible jump at the cutoff: $\hat{\tau}_Y \approx 0$ and not statistically distinguishable from zero.

Scaling a near-zero numerator by the first stage leaves a near-zero estimate:

$$\hat{\tau}_{FRD} = \frac{\hat{\tau}_Y}{\hat{\tau}_D} \approx 0.$$

Interpretation: little evidence of a sheepskin effect — the diploma on its own does not appear to raise earnings, consistent with the human-capital view. Note the IV correction matters in general: had the numerator been nonzero, the naive (uncorrected) jump would have understated the effect, and dividing by $\hat{\tau}_D < 1$ would scale it back up.

Wrap-up

What to leave with

- Fuzzy RD = imperfect compliance at a cutoff. The treatment *probability* jumps, not treatment itself.
- The cutoff indicator $\mathbf{1}\{X \geq c\}$ is an instrument for treatment received. Independence comes from local randomness; exclusion says the cutoff matters only through D.
- Estimand: $\tau_{FRD} = \tau_Y / \tau_D$ — the outcome jump rescaled by the first-stage jump.
- It identifies a **doubly local** effect: for compliers, at the cutoff. The naive sharp jump is biased toward zero.
- Defend continuity, first stage, exclusion, and monotonicity *separately*. Plot the first stage and reduced form.
- Different running variable $\Rightarrow$ different compliers $\Rightarrow$ different, separately-reported LATEs.

Questions?

References / further reading

- Imbens & Lemieux (2008). *Regression discontinuity designs: a guide to practice*. J. Econometrics 142(2). (Figs 3–4 used here.)
- Hahn, Todd & van der Klaauw (2001). Identification and estimation of treatment effects with a RD design. *Econometrica*.
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- Clark & Martorell (2014). The signaling value of a high school diploma. *J. Political Economy* (sheepskin example).
- Calonico, Cattaneo & Titiunik (2014). Robust nonparametric CIs for RD designs. *Econometrica* (`rdrobust`).
- Tilburg Science Hub, "Fuzzy RD Designs"; Facure, *Causal Inference for the Brave and True*, ch. 16.